



COMSATS University Islamabad

Department of Computer Science

Course Syllabus

Course Information

Course Code: **CSC301**

Credit Hours: **3(3,0)**

Lab Hours/Week: **0**

Course Title: **Design and Analysis of Algorithms**

Lecture Hours/Week: **3**

Pre-Requisites: **CSC211-Data Structures**

Course Contents:

This course is designed to provide knowledge of the principles and techniques used in the design and analysis of algorithms. Topics cover: Overview of Algorithm; Designing Techniques: Brute Force, Decrease and Conquer, Divide and Conquer, Transform and Conquer Technique, Dynamic Programming, Greedy Approach, Correctness of Algorithms, Analyzing Time Complexity of Iterative and Recursive Algorithms; and Computability.

Text and Reference Books

Textbook:

1. Introduction to the Design and Analysis of Algorithms, Levitin, A., Pearson, 2017.

Reference Book:

2. Introduction to Algorithms, Cormen, T. H., Leiserson, C.E., Rivest, R.L. & Stein, C., MIT Press, 2022.

Week wise Plan:

Lecture	CDF Unit #	Topics Covered	Reading Material
1.	1	The concept: What are Algorithms, Why Do You Need to Study Algorithms, What Kinds of Problems are Solved by Algorithms, Algorithms as a Technology, How to Represent an Algorithm; Properties of Algorithms: Input, Output, Precision, Finiteness, Definiteness, Correctness, and Generality.	CLRS: Ch1 Levitin: Ch1 Richard Johnsonbaugh: Ch1
2.	1	Fundamentals of Algorithmic Problem Solving: The STAIR Steps for Solving Problems, Major Factors in Designing an Algorithm.	Levitin: Ch1 Parag H. Dave: Ch2
3.	1	Iterative Algorithm Design Issues, Top-Down Design, Design using Recursion.	Parag H. Dave: Ch3
4.	9	Analysis Framework: Measuring an Input's size, Units for Measuring Running Time, Orders of Growth, Worst, Best & Average Case Behavior of Algorithms.	CLRS: Ch3 Levitin: Ch2
5.	9	Asymptotic Notations & Basic Complexity Classes: Informal Introduction, Asymptotic Notations: Big O Notation, Sigma Notation, and Theta Notation, Useful Properties involving the Asymptotic Notations, Using Limits for Comparing Order of Growth; Basic Complexity Classes i.e., Constant, Linear, Quadratic.	CLRS: Ch3 Levitin: Ch2
6.	9	Mathematical Analysis of Iterative Algorithms: Examples, and Worst, Best & Average Case Behavior of Algorithms.	Levitin: Ch2
7.	9	Analysis of Recursive Algorithms: Deriving Recurrence Relation from recursive algorithm; Solving Recurrences: Substitution Method.	CLRS: Ch4 Levitin: Ch2
8.	9	Solving Recurrences: Master Theorem for Solving Recurrences, Recurrence Tree Method.	CLRS: Ch4 Levitin: Ch2
9.	2	Brute Force Algorithms: Designing Algorithms for Sorting problem, Pattern Matching Algorithm.	Levitin: Ch3 CLRS: Ch3
10.	2	Brute Force Algorithms: Closest-Pair & Convex-Hull Problems.	Levitin: Ch3

11.	3	Decrease (by One) and Conquer: Designing Algorithm for Sorting Problem, Algorithms for Generating Combinatorial Objects	Levitin: Ch4, CLRS: Ch22
12.	3	Decrease (by One) and Conquer: Graph Traversal, Topological Sorting.	Levitin: Ch4
13.	3	Decrease (by constant factor) and Conquer: Designing Algorithm for Searching; Decrease (by variable) and Conquer: Median and Selection problem.	Levitin: Ch4 CLRS: Ch9
14.	4	Divide and Conquer (DnC): Strategy, Designing Algorithms for Sorting Problem.	Levitin: Ch5 CLRS: Ch7
15.	4	Divide and Conquer (DnC): Designing Algorithms for Closest-Pair, Convex-Hull Problem.	Levitin: Ch5
16.	4	Divide and Conquer (DnC): Designing Algorithms for Matrix Multiplication Problem.	CLRS: Ch4 Levitin: Ch5
17.	Mid Term Examination		
18.			
19.	5	Transform and Conquer: Transformation to more convenient instance, Transformation to different representation: Designing Algorithm for Sorting Problem	Levitin: Ch6
20.	5	Problem Reduction: Reduction of Optimization Problems, Reduction to Graph Problems.	
21.	6	Dynamic Programming: Comparison of DP, Divide & Conquer, and Elements of Dynamic Programming.	CLRS: Ch14 Levitin: Ch8
22.	6	Dynamic Programming (Edit Distance Problem): Edit, Edit Distance Designing DP Algorithm for Edit Distance Problem & its Time Complexity, and Applications Analysis of DP Edit Distance. Designing DP Algorithm for LCS	CLRS: Ch14
23.	6	Dynamic Programming (0/1 Knapsack Problem): Problem Analysis, Notations, Designing DP Algorithm for 0/1 Knapsack Problem, and Applications of 0/1 Knapsack Problem.	CLRS: Ch14 Levitin: Ch8
24.	6	Dynamic Programming (Matrix Chain Multiplication): Problem Analysis, Notations, Designing DP Algorithm for MCM, and Applications of MCM.	CLRS: Ch14
25.	7	Greedy Algorithms: Elements of the greedy strategy, Designing Algorithm for Data Compression (Huffman Tree and code), Activity Scheduling	CLRS: 15 Levitin: Ch9
26.	7	Greedy Algorithms: Designing Algorithm for activity-selection problem	CLRS: Ch15
27.	8	Correctness of Iterative Algorithm: Pre-condition, Post-condition, Loop Invariants.	Thomas: Ch1, Ch4,
28.	8	Correctness of Iterative Algorithm: Quick Review of Mathematical Induction; Initialization, Maintenance, and Termination, and Illustrative Examples.	Thomas: Ch1, Ch4
29.	8	Correctness of Recursive Algorithm: Proving Correctness of Recursive Algorithm using Induction, and Illustrative Examples.	Thomas: Ch5
30.	10	Introduction to Complexity Classes & Computability: Polynomial time Polynomial-time verification	CLRS: Ch34 Levitin: Ch11
31.	10	NP-completeness and reducibility	CLRS: Ch34 Levitin: Ch11
32.	10	NP-completeness proofs, NP-complete problems	CLRS: Ch34 Levitin: Ch11
Graduate Attributes (GAs)			

GA #	Description
2	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
3	Identify, formulate, research literature, and solve <i>complex</i> computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
4	Design and evaluate solutions for <i>complex</i> computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

Course Learning Outcomes (CLO)

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	GA
CLO-1	1-2	Demonstrate fundamental concepts of algorithm along with an algorithmic approach to a given problem.	<i>Understanding</i>	2
CLO-2	3-7	Design new algorithms for different computational problems.	<i>Creating</i>	2,4
CLO-3	8	Prove correctness of an algorithm using loop invariant and induction.	<i>Applying</i>	2,3
CLO-4	9	Analyze best, average, and worst-case behaviors of an algorithm.	<i>Analyzing</i>	3
CLO-5	10	Explain the concept of various complexity classes with examples.	<i>Understanding</i>	2

CLO Assessment Mechanism

Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4	
Assignments		Assignment 1&2	Assignment 3	Assignment 4	
Mid Term Exam	Mid Term Exam	Mid Term Exam	-	-	
Final Term Exam	Final Term Exam				

Policy & Procedures

- **Attendance Policy:** Every student must attend 80% of the lectures as well as laboratory in this course. The students falling short of required percentage of attendance of lectures/laboratory work, is not allowed to appear in the terminal examination.

- **Course Assessment:**

	Quizzes	Assignments	Mid Term Exam	Terminal Exam	Final Marks
Theory (T)	15	10	25	50	100

- **Grading Policy:** The minimum passing marks for each course is 50% (In case of LAB; in addition to

theory, student is also required to obtain 50% marks in the lab to pass the course). The correspondence between letter grades, credit points, and percentage marks at CUI is as follows:

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	F
Marks	>= 85	80 - 84	75 - 79	71 - 74	68 - 70	64 - 67	61 - 63	58 - 60	54 - 57	50-53	< 50
Cr. Point	3.67-4.00	3.34-3.66	3.01-3.33	2.67-3.00	2.34-2.66	2.01-2.33	1.67-2.00	1.31-1.66	1.01-1.30	0.10-1.00	0.00

- **Missing Exam:** No makeup exam will be given for final exam under any circumstance. When a student misses the mid-term exam for a legitimate reason (such as medical emergencies), his grade for this exam will be determined based on the Department policy. Further, the student must provide an official excuse within one week of the missed exam.
- **Academic Integrity:** All CUI policies regarding ethics apply to this course. The students are advised to discuss their grievances/problems with their counsellors or course instructor in a respectful manner.
- **Plagiarism Policy:** Plagiarism, copying and any other dishonest behaviour is prohibited by the rules and regulations of CUI. Violators will face serious consequences.